



***CP2411: 3D Modelling and
Animation***

Assessment Part 3

Project Title: "Slacker to Soldier"

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Part 1

1. Story background

Storytelling: Subtitles and audio will be used in the final video.

Slacker to Soldier

Once upon a time, in a disciplined army unit, there was one soldier who was always lazy, irresponsible, and constantly caused trouble for his teammates. His name is Lim.

During an overseas training in Thailand, something unexpected happened. On the first night in Thailand before the first outfield mission, the necklace (amulet) from his father, which was meant to keep him safe, slowly flew up, allowing him to visit a mysterious angel in his dream. The angel asked, "Why are you so careless and irresponsible? Why do you ignore your duties?" Before the soldier could answer, the angel touched his forehead and said, "Let this gift show you your true potential." The necklace on his neck changes its appearance, becomes two bracelets on his hand, and the light comes out from it, waking him up.

He woke up in his bunk, unsure if what he experienced was just a dream, but when he checked the necklace on his neck, he found out his necklace looked different; it seemed like it was not just a dream.

When he was thinking about the dream, a bunk spot check from his sergeant and commander broke his thoughts, and once again, the whole team was punished when the commander saw his untidy bed. After the punishment, the sergeant asked them to be ready in 5 minutes. At the same time, the scolding voices of his teammates reminded him, he was still the team's biggest slacker. After loading their equipment onto the tonner, they took it to a hill deep in the forest, where they had to set up defensive structures.

As usual, he began to slack off. But something strange happened—when he lifted the heavy equipment, it felt unusually light. Curious, he carried more... and it still felt effortless. Inspired, he took on even more responsibility, helping the team set up defences in record time. The squad noticed. Whispers turned into cheers. They were

impressed but confused by his sudden strength and motivation.

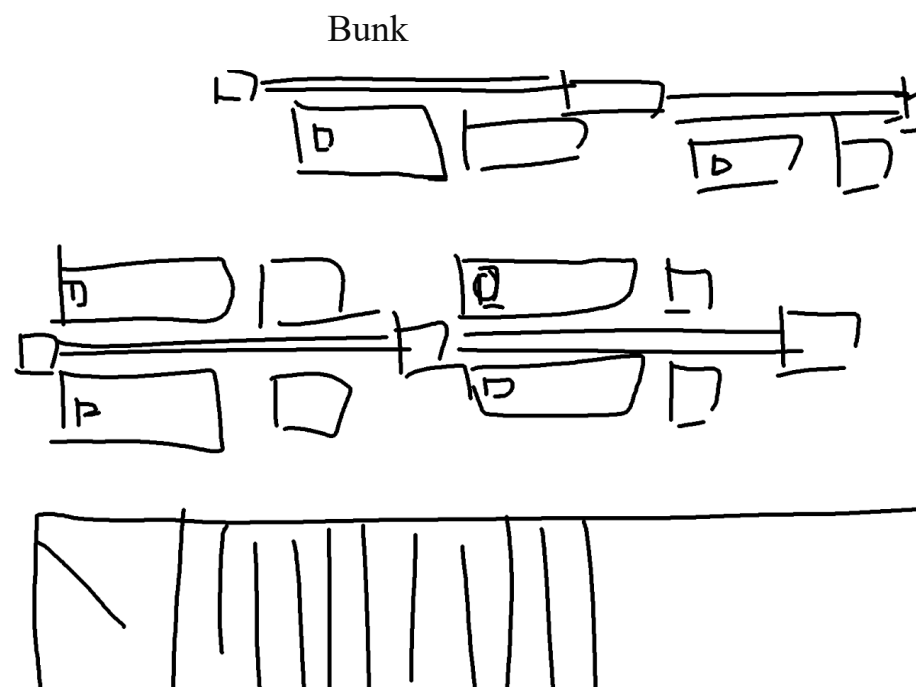
Later that day, the team engaged in a simulated combat exercise. To everyone's surprise, the once-lazy soldier fought fiercely and skillfully, taking down many opponents. His efforts were a major reason for their victory.

When they returned to base, the commander announced something unheard of; it was the first time the defence team had ever beaten the attackers in this scenario. To celebrate, he treated the whole squad to beer.

That night, the soldier sat quietly, watching his teammates laugh and share stories with him, not behind his back, but beside him. For the first time, he felt something different: belonging. He smiled, knowing the angel's power wasn't just physical strength. It was the chance to change.

From that day on, he worked harder, trained seriously, and became a true team player. By the time he graduated from the army, he was no longer a lazy soldier. He had become a better man—respected, reliable, and proud.

2. Sketches - Design pictures



Bunk Story: This will be the place main character wakes up after meeting the angel.

The main character is a slacker, the bed and cabinet will be messy compared to others, the scolding part by the commander will also be here also and hold team member is

punished to do push-ups because of the main character's slackness.

Heaven is like a magical place.



Angel Scene: This will be a simple place where only the “Angel” and the main character are in a sky realm/sky mirror place.

Story: This is the place where the main character gets his magical power to unlock his full potential.

Outfield (forest)

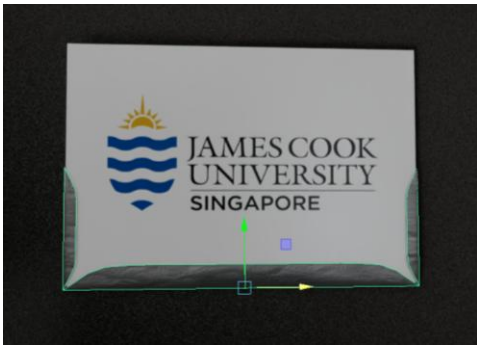


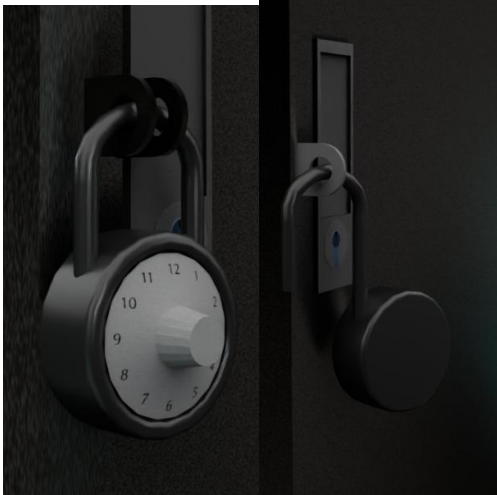
Outfield Scene: This will be the part where the main character finds his potential

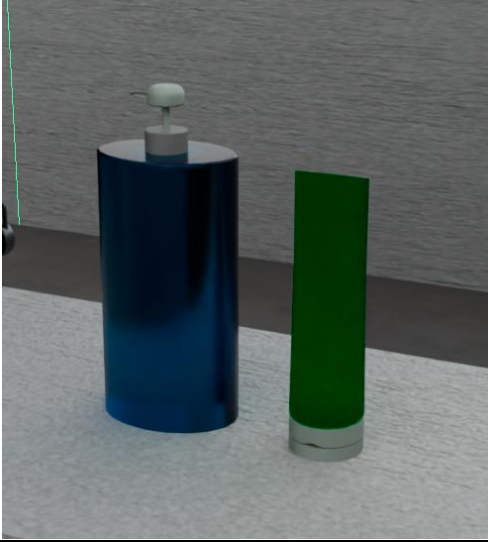
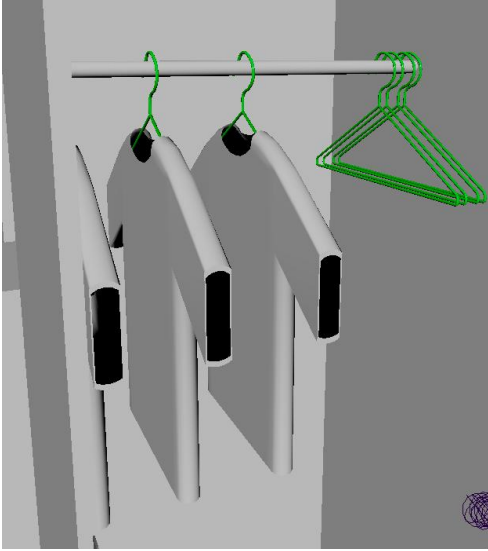
given by the angel and helps the team to set up the defences and win the gun fight.

3. 3D object list

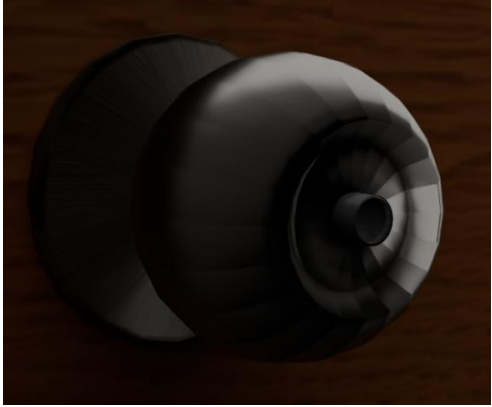
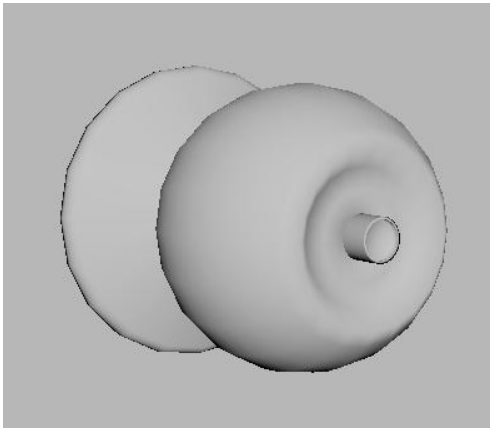
<i>3D Object List</i>	<i>Screenshot</i>	<i>Explanation</i>
Towel		What: Army green towels for soldiers to dry themselves. Why: There are two towels, one tidy and one untidy, to differentiate that the main character is too lazy to put the towel nicely. How: Created using cube at start with the use of extrude and rotate to create a curve, and bevel to make the edge smoother and round, and finally mirror.
Bed, Pillow, Blanket		What: A Place for soldiers to sleep and rest. Why: This is where the main character sleeps and dreams. How: The bed and pillow are a cube that scales into a rectangle and applies a bevel to all edges. The blanket is an extent using extrude of

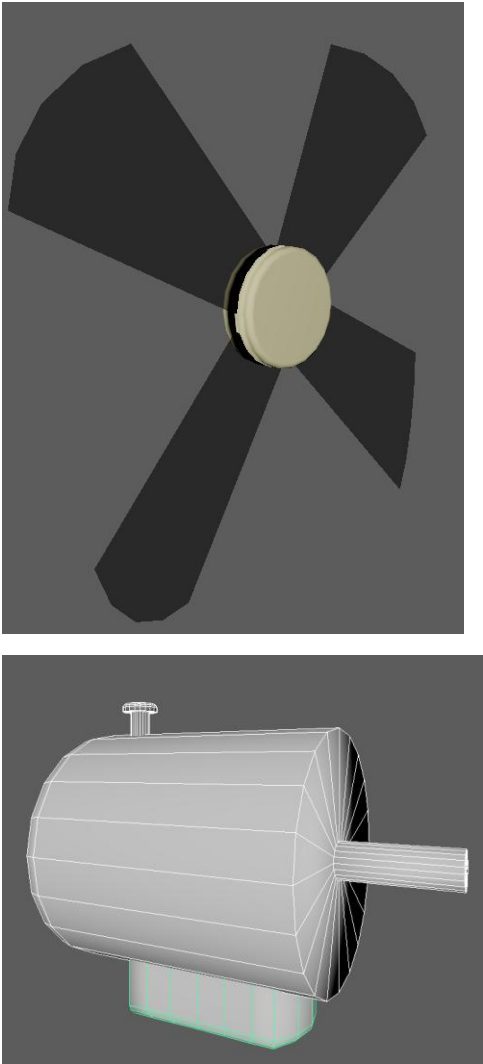
		the towel.
Bed Frame		What: A Frame to place the bed Why: More realistic decoration How: Made by multiple cylinders and cubes. The bending part is done by extrude, rotate and mirror.
Cabinet		What: A container to store army equipment and clothes. Why: The difference in the main character's unlocked locker shows the lack of security of his stuff. How: The Body is a cube with an added loop edge at each side, extruded to make the deep, and an edge loop is added in the middle to extrude out for the middle layer. A cube door on the front and multiple cylinders on the door side.
Name tag and frame		What: A name tag that shows the soldier's name. Why: Here, it would be the JCU logo, and the main character will be my JCU ID. How: The main tag itself is just a plane. The frame is also a

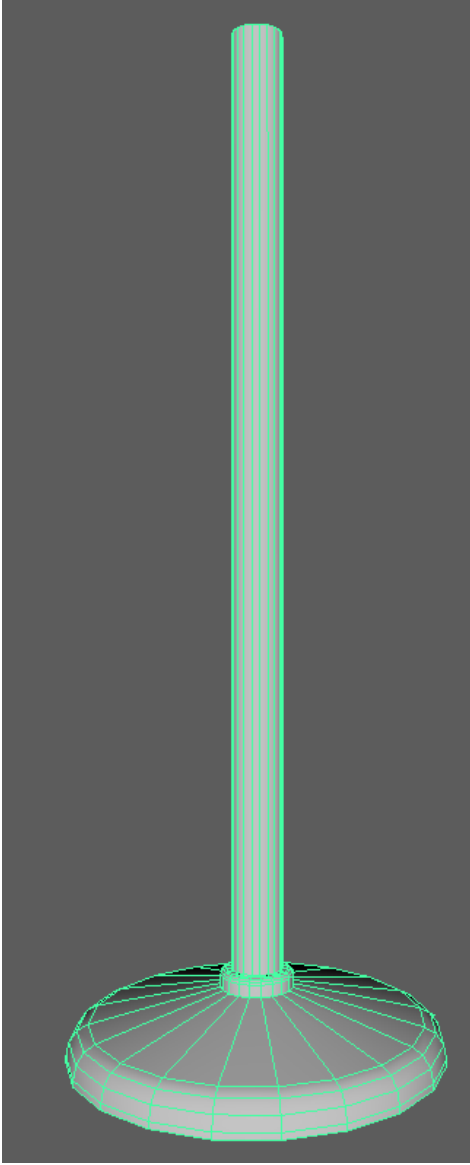
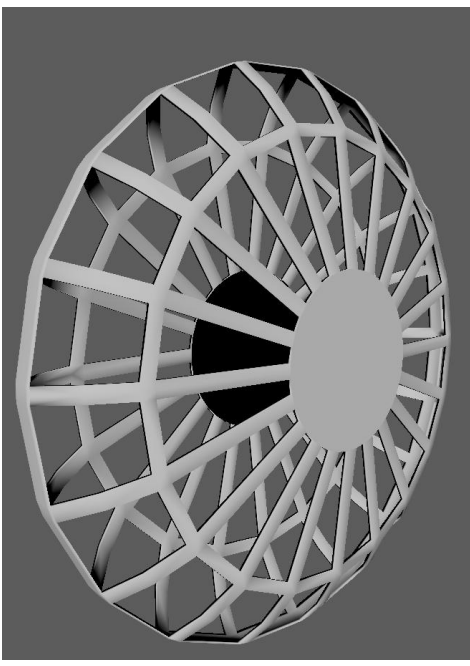
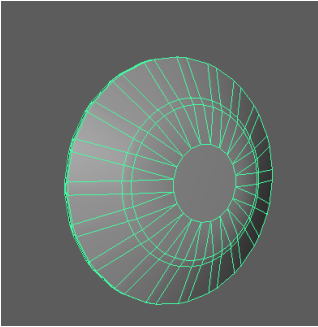
		frame, but with extrude 90 degrees up and extrude out with a bevel on the vertex to make it look nice.
Lock		What: Number lock to lock the cabinet Why: Details for the unlock/lock cabinet How: Create by 3 cylinders, top extrude and bend, body extrude inward and bevel, and number part add loop edge and scale.
Cylinder light		What: Cylinder light to light the bunk Why: Decoration to make the lighting more logical. How: A Light bulb is a cylinder. The rack is a cube extended and added edge loop on both sides, and extruded out with some adjustment on the vertex to make it look round.

Shampoo and facial		What: Small item for decoration Why: To show that the main character always places his stuff anyway How: All components are created by cylinders and scaling the vertex to make different sizes and shapes.
T-shirt Hanger		What: An inner view of the cabinet for the main character to unlock the cabinet. Why: Decoration of the inner cabinet How: Hanger is made by extruding a cylinder slowly and bending it, finally, mirror. A T-shirt is created by first creating a cube and by moving the vertices and deleting faces to show the hole.

T-shirt outside		What: A T-shirt that is thrown on the outside. Why: More items to show the slackness of the main character. How: A copy of the T-shirt is inside the cabinet. Adding multiple edge loops in the middle makes it bendable.
Door Door frame		What: The entrance to the bunk Why: This is for the sergeant to come in for bunk check and for the character to move out to the outfield. How: The Door is a scaled cube, and the frame is a cube with an added loop edge, a deleted face in the middle, bridging the hole created after deleting and bevelled the inner side.

Door handle	 	What: The door handles open or close the door. Why: A decoration that shows a traditional design army bunk door. How: It is created by initially creating a sphere by moving inwards to create the front and deleting a hole for the button. Backside is by scaling the vertex to make it flat, extruding inward and outward again to make the middle-connected part, and finally scaling the vertex out to make the plate-like stuff on the door. The button in the middle is just a cylinder with a connected edge (to make a new circle) and extruded inward. It looks smooth because of the softened edge.
Window, Frame		What: A Window that can be opened Why: Decoration and for the lighting to come in through the transparent window. How: The Window is a cube with a transparent texture, and the frame is quite the same as how I created the door frame.

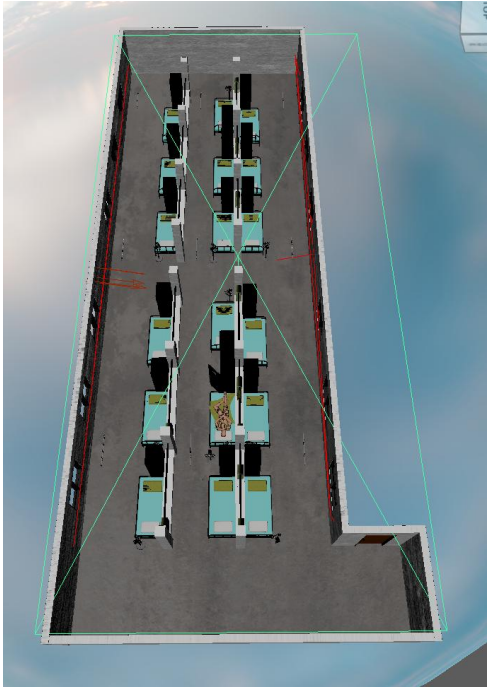
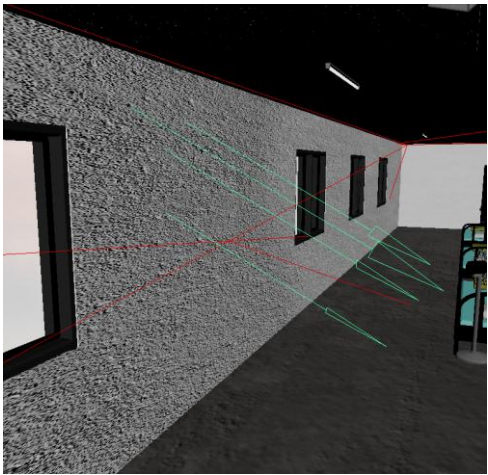
Window handle		What: Handle to ensure that the window closes nicely Why: Decoration on the window How: It is originally a cube, by extruding three sides and scaling the vertices to make the extruded side thin.
Stand Fan 		What: A Stand fan is the main cooling method in the bunk. Why: The Army bunk, which didn't have an air conditioner, made it the main method for soldiers to stay cool. How: Fan Blade: A cylinder was added with an edge loop in the middle and extruded out the selected face to make the blade. After adjusting the size and shape, I rotated the blade to simulate the real work blade. Motor part: the main body is a cylinder with one side bevelled and the other side connected edge, and extrudes out. The bottom part is a cylinder that adjusts the middle vertex and scales it to connect to the base. Base: It is also a cylinder with multiple extrude and

	 	scaling. Fan net casing: It is created by a cylinder with some bevel on the edge and vertex. After adjusting the edge, I deleted faces to represent the net and bridged the inner part (to prevent black inner texture).  I recreated a sample (not the original) before the face was deleted to show how I did it.
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Human		What: This will be the model for the two main characters. Why: This will be the main character, and I plan to make this for two main characters using this model only, and for others, I plan to make a very low-poly model for them How: This is made by a tutorial from YouTube, the link can be found in the reference section. I also added the recording of how I did with fast-forward time elapsed.
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4. Lighting

Sky Dome Light		What: Main sky light with the sky texture. Why: For the main lighting of the sky and a more realistic sky when seen out of the window. How: The texture is downloaded online and scaled until it covers the interior environment.
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3 Area light		What: Ceiling (top), window (left, right) lighting to light up the interior space. Why: As the sky dome light cannot fill the interior, the area light is used to represent the light that comes from the window and the cylinder light. How: By adding Arnold lighting and scale until it fits the area needed.
Directional Light		What: Directional light from the window Why: To represent the light from the sun in one direction, and it will produce some realistic shadow as needed. How: By adding Arnold Direction Lighting and adjusting the rotation until the desired effect is achieved.

5. 3D game world

3D Environment: Army bunk (interior scene)

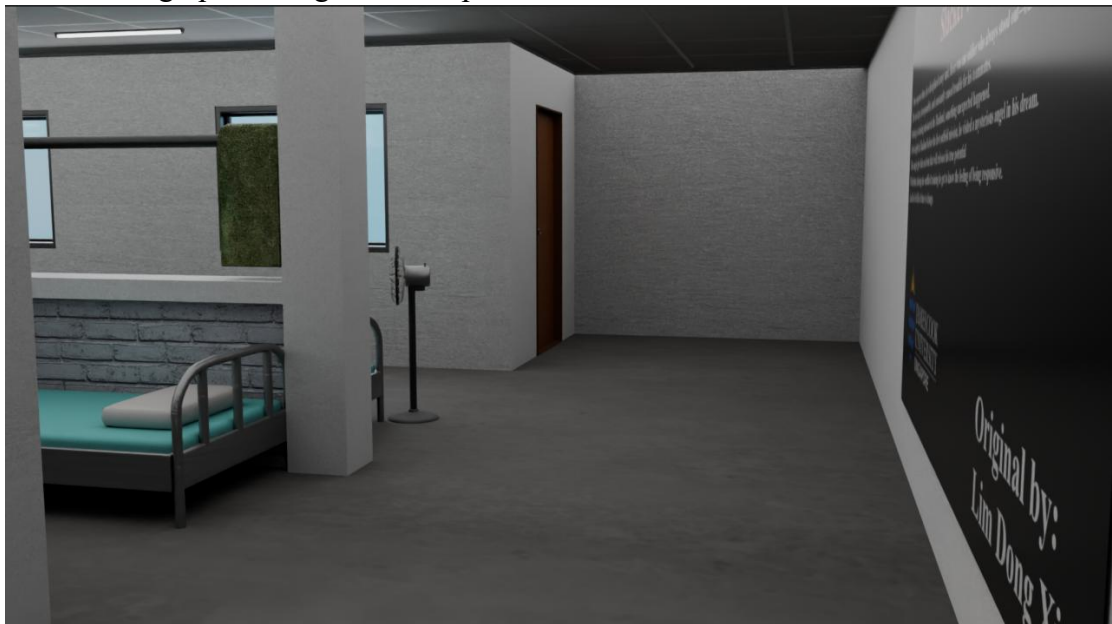
Animation plan:

- Ring float
- Wake up
- Sergeant spot check
- Get punished

First part: Lim sleeps, and the necklace floats



After waking up, the sergeant will spot check the bunk entrance from the entrance.



6. Story character

The main character in the story: Lim

The second important character will be the main character, Buddy, using the same model named Jackie.

The story will be mostly in first-person view in the main character's head, so it won't be hard to differentiate them. Other characters will be using a very low-poly model.



7. Texture

Texture-wise wise I texture everything in the object list. Mostly, I have done it using Photoshop to edit pictures, adjusting the 3D Substance Painter built-in material and Normal mapping tool to create a normal map using an image.

8. Video file

10-second scene showcase backup link:

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[pZXciOiJTaGFyZURpYWxvZy1MaW5rliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXcifX0%3D](https://myicuedu-my.sharepoint.com/:v:/g/personal/dongxian_lim_my_jcu_edu_au/EZSmRXT3JmtIg082S59zD2UB_DQTUmqGf1CLKGPLBYJDdg?e=MrgBZ9&nav=evJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJTdHJIYW1XZWJBcHAiLCJyZWZlcnJhbFZpZXciOiJTaGFyZURpYWxvZy1MaW5rliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXcifX0%3D)

Progress Video (Evident video)

Human body and clothes (This video is quite long so I didn't upload the original mp4 to the dropbox): [https://myicuedu-](https://myicuedu-my.sharepoint.com/:v:/g/personal/dongxian_lim_my_jcu_edu_au/EZSmRXT3JmtIg082S59zD2UB_DQTUmqGf1CLKGPLBYJDdg?e=MrgBZ9&nav=evJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJTdHJIYW1XZWJBcHAiLCJyZWZlcnJhbFZpZXciOiJTaGFyZURpYWxvZy1MaW5rliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXcifX0%3D)

[my.sharepoint.com/:v:/g/personal/dongxian_lim_my_jcu_edu_au/EZSmRXT3JmtIg082S59zD2UB_DQTUmqGf1CLKGPLBYJDdg?e=MrgBZ9&nav=evJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJTdHJIYW1XZWJBcHAiLCJyZWZlcnJhbFZpZXciOiJTaGFyZURpYWxvZy1MaW5rliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXcifX0%3D](https://myicuedu-my.sharepoint.com/:v:/g/personal/dongxian_lim_my_jcu_edu_au/EZSmRXT3JmtIg082S59zD2UB_DQTUmqGf1CLKGPLBYJDdg?e=MrgBZ9&nav=evJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJTdHJIYW1XZWJBcHAiLCJyZWZlcnJhbFZpZXciOiJTaGFyZURpYWxvZy1MaW5rliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXcifX0%3D)

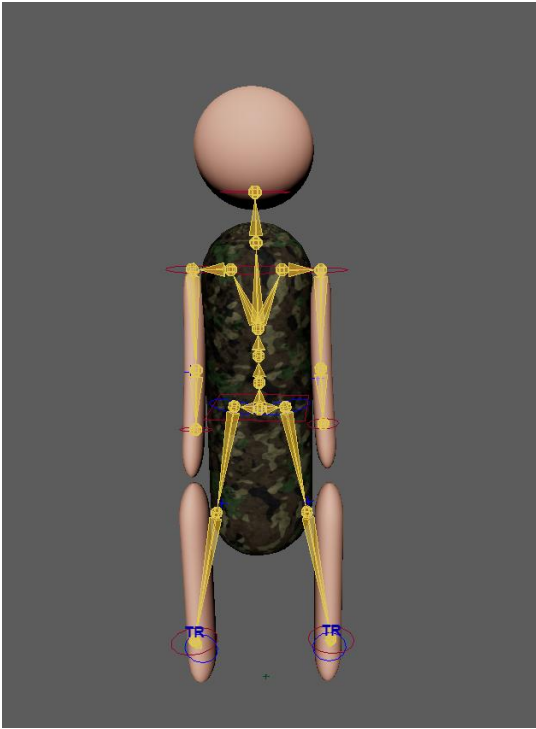
9. Part 2

9.1 Update content

9.1.1 Update item list

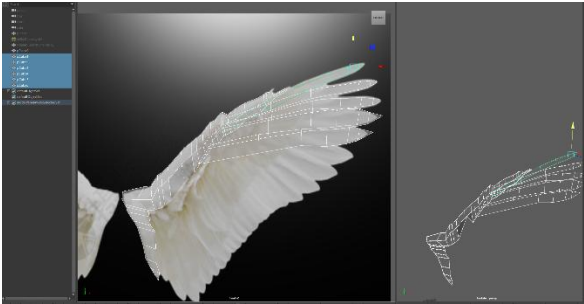
Update the bunk scene 3D object list.

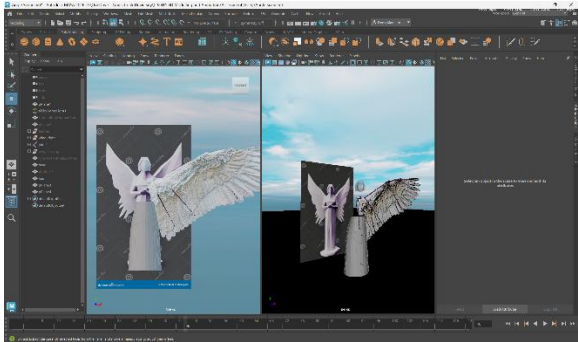
Necklace with rune ring		What: A simple chain worn around the character's neck, with the original rune ring given by his father. Why: The main reason the character can visit the angel. How: I created it the same as how the rune ring is done, but with an adjustment on the vertex to make it go around the character's neck.
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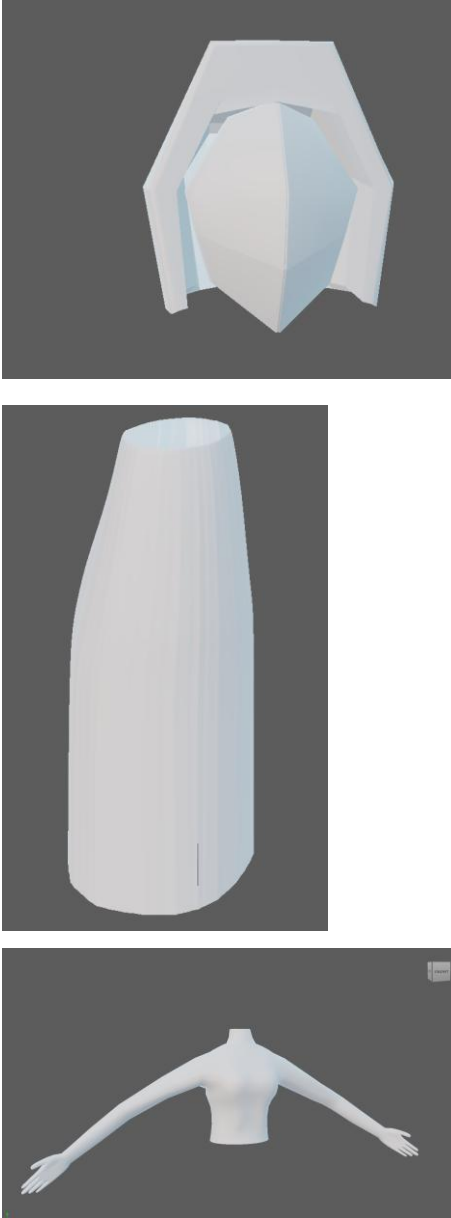
Other character		What: A simplified humanoid figure made of basic, smoothed shapes like spheres and cylinders. Why: Used to populate the scene efficiently without taking up heavy computing resources, while keeping focus on the 2 main characters' detail and animation. How: Modelled using basic sphere for the head, smoothed cylinders for the body and scaled for the hand and legs and positioned to form a humanoid silhouette with minimal detail.
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9.1.2 New scene object list

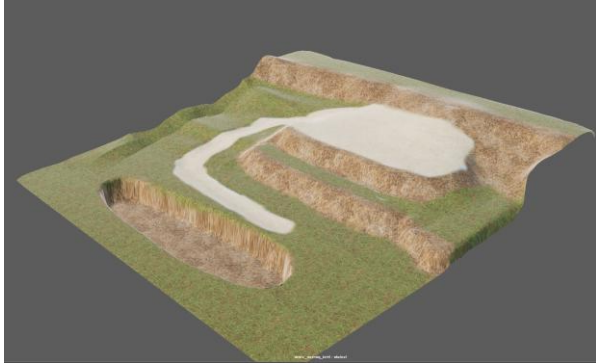
New 3D object list (Angel Scene)

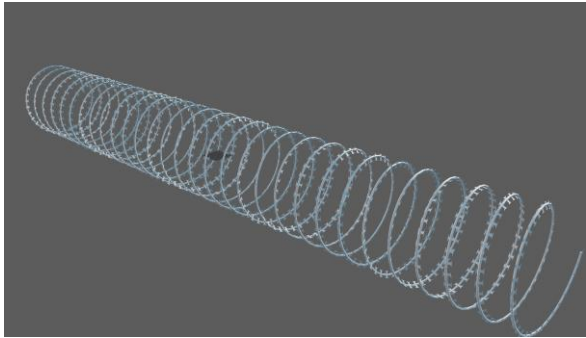
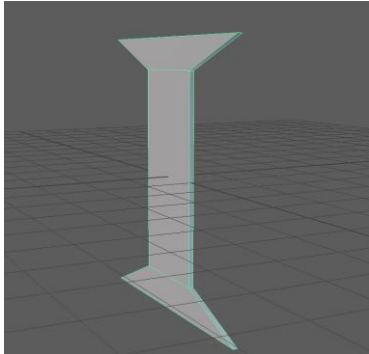
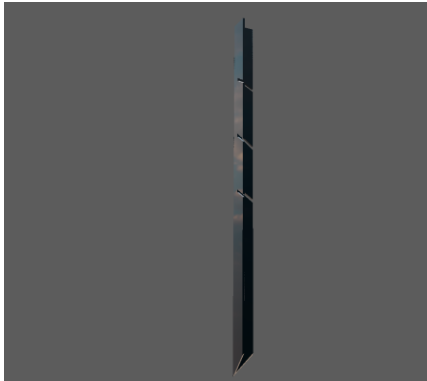
Wing	 Progress pic (half way): 	What: A pair of wings resembling angel feathers, attached to the angel's back. Why: To emphasize the character's divine or heroic nature and visually enhance their fantasy-themed design. How: I try to model it in one object at first which didn't work out. I get the idea from the YouTube video of wing rigging
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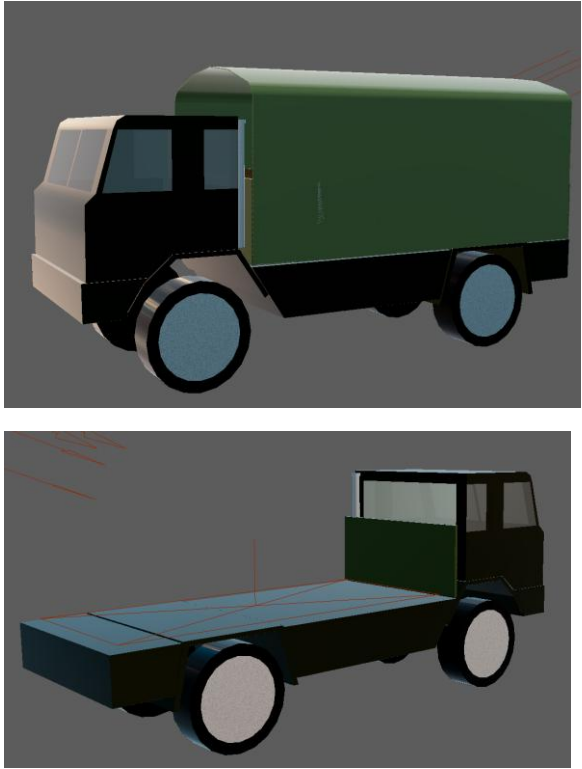
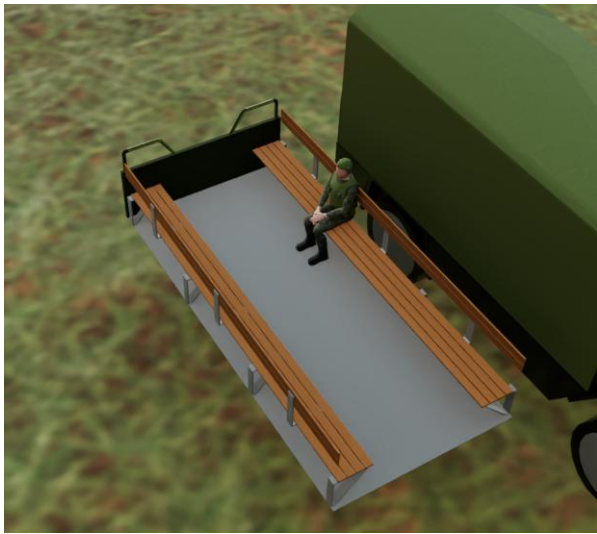
		that separate each feather and the bone part of the wing using smoothed and extrude cube .
Angel floating ring		What: A decorative floating golden ring positioned above the head, commonly associated with angels. Why: To reinforce the character's celestial or divine identity. How: Simply created by cylinder after deleting the face in middle and connect the inner part.
Angel	 Angel progress (half): 	What: The main humanoid figure that forms the core structure of the angel character. Why: The base for attaching all other elements (wings, ring, clothes), allowing for future rigging and animation. How: The main idea of the model is base on the picture from webpage Dreams Time: Low poly angle Head and hair: I

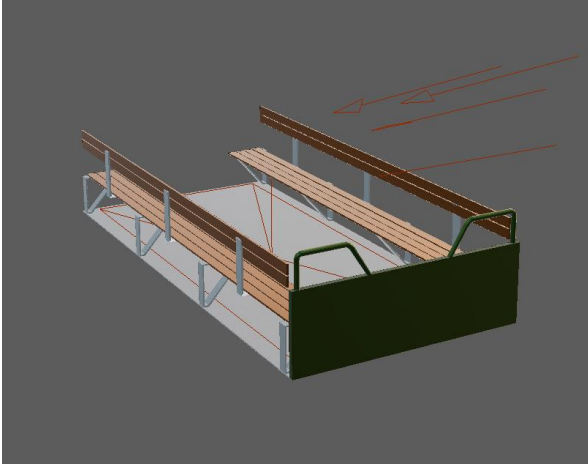
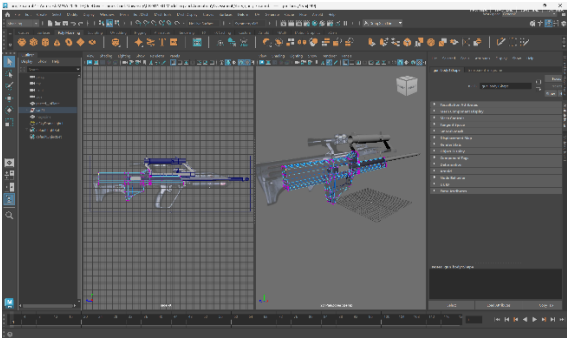
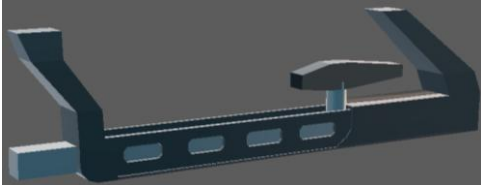
		make it low poly reference from the low poly angel picture. Both of them are created from a cube. Skirt: Started with a plane and used multiple extrusions to create wrinkles. Adjusted the waist and butt for realism, added a central edge, and mirrored the model for symmetry. Angel body/shirt: Modelled the body and arms using the same technique as the main character; fingers and palm were copied from the main character.
Rune Bracelet		What: A circular on the character's wrist, originally a necklace ring adapted into a bracelet. Why: Decoration to the character, representing mystical or divine power due to its runic design. How: Created same way as the angel ring.

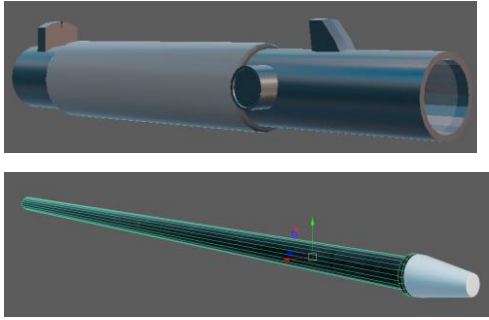
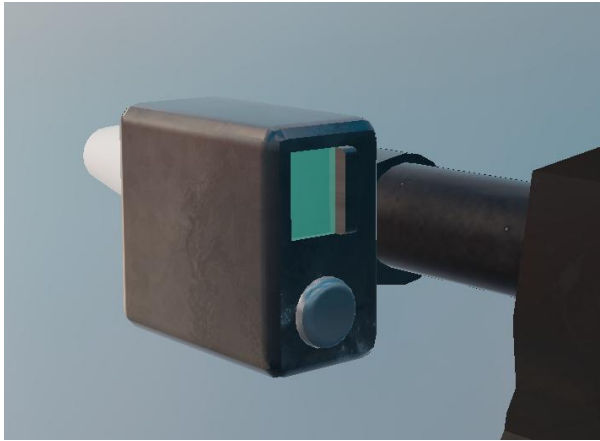
New jungle scene 3D object list

Tree (Trunk/ Leaves)		What: A stylised low-poly tree consisting of a trunk, a few branches, and spherical foliage. Why: To populate the forest environment and create a natural, lively setting. How: Trunk: Made from a cylinder by adjusting vertices, then extruding and scaling to form the trunk and branches. Leaves: Learned by watching a tutorial from YouTube.
Terrain		What: A large landscape mesh that serves as the ground for the scene. Why: Provides the environment where the characters and props are placed. How: Made by sculpting a subdivided plane to form paths and hills. Texture is created using multiple masks and material from 3D substance.

Thorn Wire	 	What: A coiled wire-like barrier with thorns, like a waist-height spring. Why: Used as a defensive structure by soldiers to block or slow the movement of the enemy. How: Made using a Helix for the coil and modified cubes distributed with MASH to form the thorns.
Iron Bar	 	What: A metal stake driven into the ground to anchor the thorn wire. Why: Provides stability and holds the thorn wire in place during setup on rough terrain. How: Modelled from a cube with adjusted vertices to create a sharp base and a hole near the middle for securing the wire.

Tonner		What: A military transport truck used to carry soldiers into the forest. Why: For troop mobility and deployment in rough terrain. How: Tonner Body: Built using multiple modified cubes for a low-poly structure. Wheels: Simply created using scaled cylinders to resemble tires. Outer Cargo: Formed from a cube with front and back faces removed to create space. Edges were extruded and connected to form inner surfaces, with top parts adjusted for a more realistic look.
Tonner Cargo (inner)		What: The inner section of the transport vehicle, where characters sit during the ride. Why: Serves as the setting for a cutscene where the character wakes up inside the tonner, adding narrative immersion. How: Benches: Benches on both sides are made

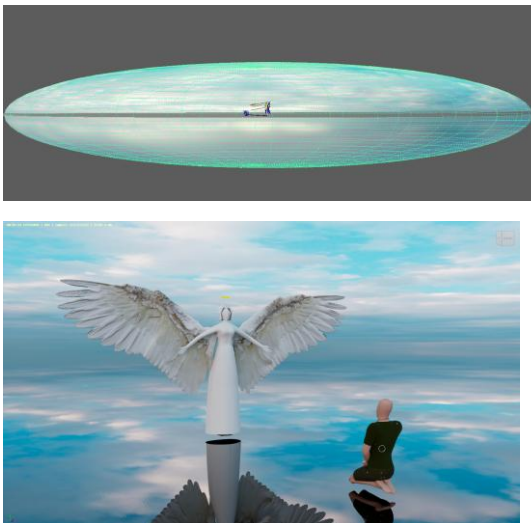
		from multiple scaled rectangular cubes. Benches' bottom: The bench supports are created using cubes with added edges, extruded and connected to form triangle-like structures. Tonner back door: The back door area features a bent tube on top and a scaled cube to complete the design.
Sar 21	Model progress:   	What: Standard-issue rifle used by Singapore military forces. Why: Represents the weapon used during training, with a modified muzzle to reflect the use of blank rounds for safety and realism. How: All components were modelled by extruding and adjusting cubes and cylinders based on reference images. Gun body: Formed using three modified cubes. Bolt: Built with two cubes and a connecting cylinder. Scope: Created using

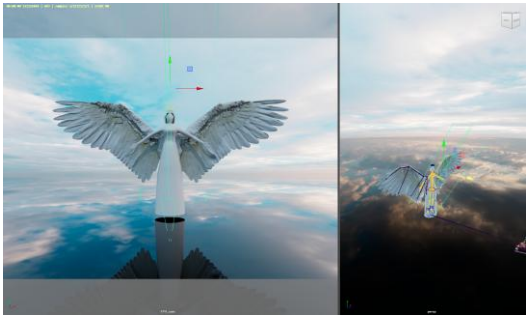
		two modified cylinders with extrusions to form near-sight details. Muzzle and Barrel: Made from a single cylinder with extruded edge details and a conical muzzle shape at the tip.
Lazer tech (tip of the gun)		What: A device mounted at the muzzle of the rifle that simulates gunfire using lasers. Why: Used in military training to safely simulate firing; works with soldier-worn laser sensors to detect hits without using live ammunition. How: Built using several modified cylinders and cubes to represent the real-world laser module.

Soldier with an outfield equipped		What: Character model dressed in standard outfield military gear. Why: Represents a soldier in field training, helping set the scene of a military environment with realistic gear. How: Equipment such as the helmet, vest, long pants, and long shirt is modelled using the same technique as the bunk character's clothing. Extra pouches on the vest are made from bevelled cubes to give them a soft, realistic appearance. Boots are created using the idea of the character's foot, using a cube initially.
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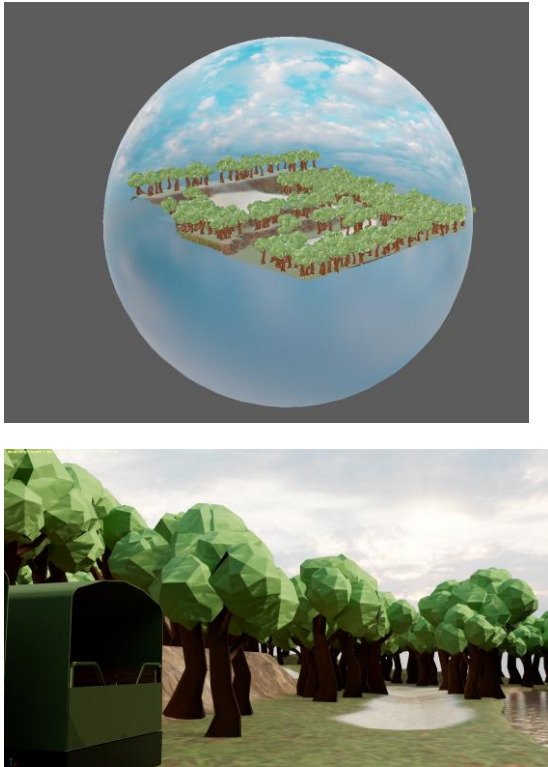
9.1.3 Update lighting List

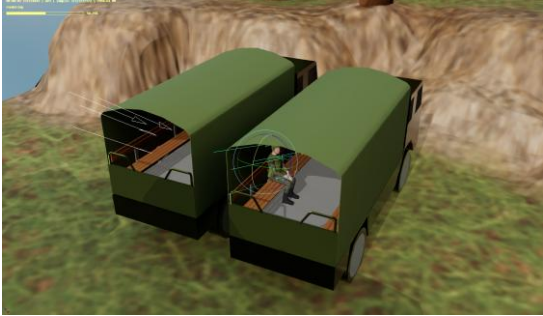
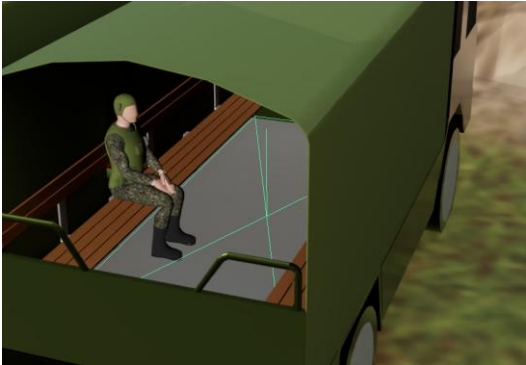
Angel Scene

Ai Sky Dome Light		What: A dome-shaped HDRI light that casts ambient illumination across the entire scene using a sky image. Why: Used to simulate realistic outdoor lighting and reflections, especially to replicate the Tokyo Ghoul opening-style effect with the sky reflection on the ground mirror.
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		How: Light is set up using an HDRI sky texture. It scaled to surrounds the entire scene and evenly emits light, allowing reflective surfaces like the ground mirror to display the sky image, enhancing realism and atmosphere.
Directional light		What: A parallel light source that simulates sunlight from a direction. Why: Used mainly to illuminate the angel wings clearly, ensuring their details are not lost in shadows or low lighting. How: Positioned and angled to highlight the wings.

Forest

Ai Sky Dome Light		What: A dome-shaped light source that provides ambient environmental lighting using an HDRI. Why: Used to simulate natural outdoor lighting across the entire forest scene. How: Applied an HDRI image of a natural sky to the dome, allowing it to wrap light around the terrain and objects for a more realistic and immersive look.
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Two directional lights		What: Parallel light sources that simulate sunlight and cast consistent shadows across the scene. Why: - The first light is used to cast strong and natural-looking shadows for realism in the environment. - The second light is placed to enhance lighting inside the tonner, ensuring the characters and props are visible despite the enclosed space. How: One positioned to simulate sunlight direction for realistic shadows, while the other is adjusted to target and brighten the tonner's interior.
Area light		What: A rectangular light source that emits light from a flat surface in a specific direction. Why: Added to enhance bright inside the tonner where the Sky Dome Light alone was insufficient, especially for darker or enclosed areas. How: Positioned in tonner's interior with intensity and direction adjusted.

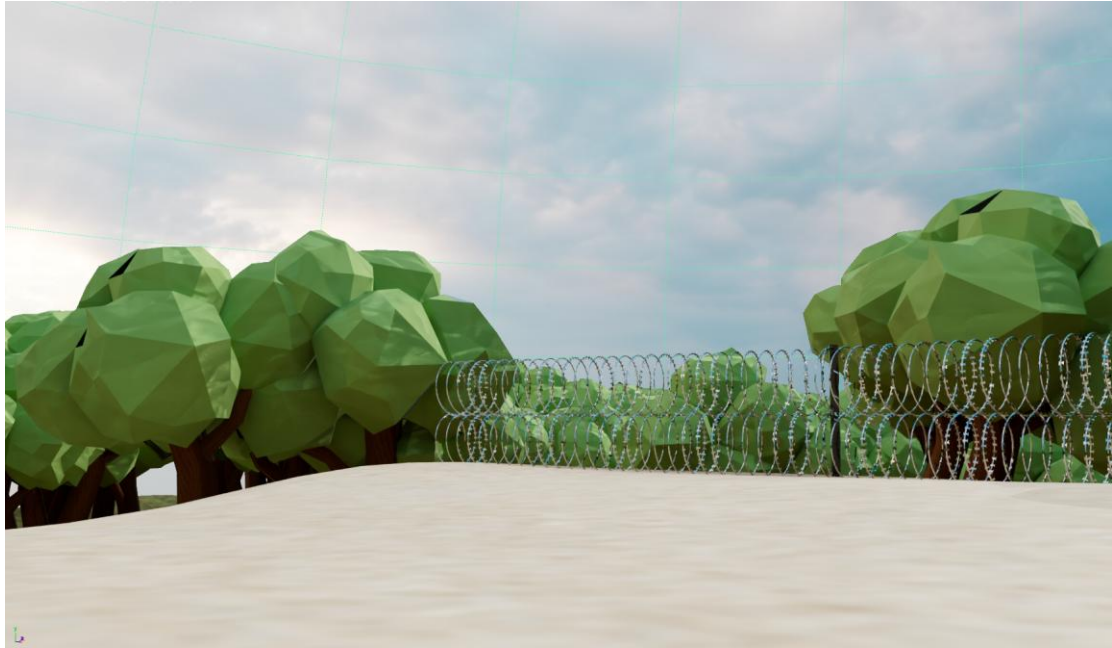
9.1.4 Update 3D Game World

Angel Scene



Forest Scene





I created the forest with my tree model using MASH following the YouTube tutorial video from Sam Biddle (2021, April 2). And I learn how to create mask for MASH to decide tree placement learn from (Pisco, the 3D guy, 2025) video.

9.1.5 Update Story character

Main Character (Lim and Jackie):



For clarification, the first skeleton (the sleeping Lim) is created using Actor Core

in the bunk during the angel scene, as I wanted to represent my character sleeping in the bunk. As I wanted to do these two scenes in first-person view, so the rigging is not really important, I used the Actor Core rig instead for Lim in these two scenes.

But I found out that the Actor Core wasn't as good. So to learn how to rig a character in Maya, all other character is created using Maya Quick Rig, and I also added finger rigging for Lim and Jackie using joints (I created a video on how I rig fingers) for future finger movement needed.

Other Character:

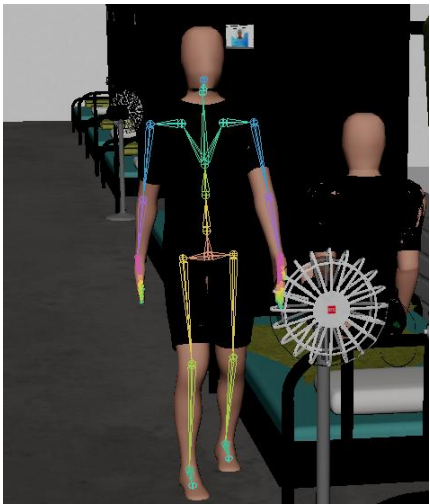
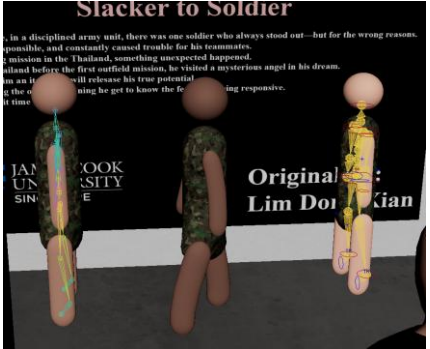
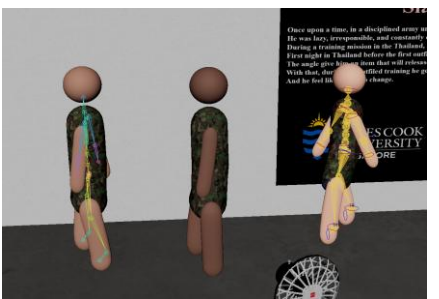
Other characters will be represented with a low-poly character with a simple, quick rig to speed up the animation and render sequence progress.



9.2 Animation

9.2.1 Movement

Bunk scene

Jackie and Lim's movement			What: Character movement animation for Jackie and Lim, including walking and turning. Why: To bring the characters to life and show their movement during key scenes. How: Animated using keyframe animation in Maya. A base movement was created and layered with separate rigging animation.
Sergeant and commander movement			What: Simple object-based movement and rotation for the Sergeant and Commander. Why: To show their presence and interaction in the scene. How: Movement was done using keyframes on the object level. Position and rotation were animated to simulate walking and turning.

Necklaces float		What: Animation of the necklace floating upward on its own. Why: To symbolise the beginning of a magical event and the angel's appearance in the dream, marking a turning point in the story. How: Keyframe animation was used to move and rotate the necklace slowly upward with a slight floating motion. Glow and particle effects were added to enhance the magical feeling.
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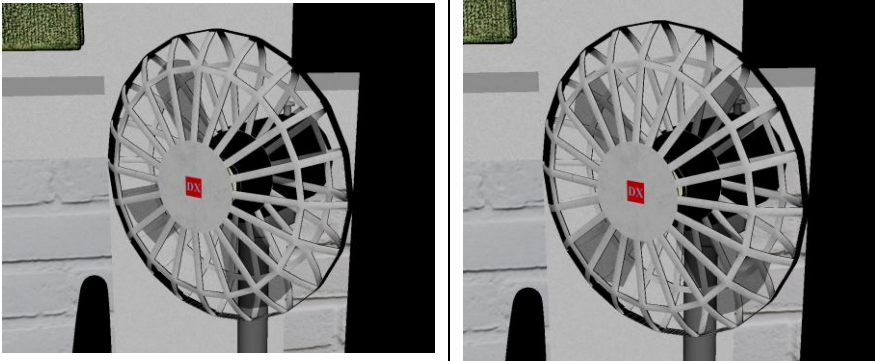
Angel scene

Angel, wing and angel ring movement loop		What: Looping animation for the angel character, floating wings, and rotating angel ring. Why: To create a calm, divine, and magical presence during the dream scene, showing that the angel is not a normal human figure. How: The angel's body gently moves up and down in a floating loop. Wings have a subtle flapping motion, and the angel ring floats slowly above the head. All movements are done using keyframes in a layer and set to loop for
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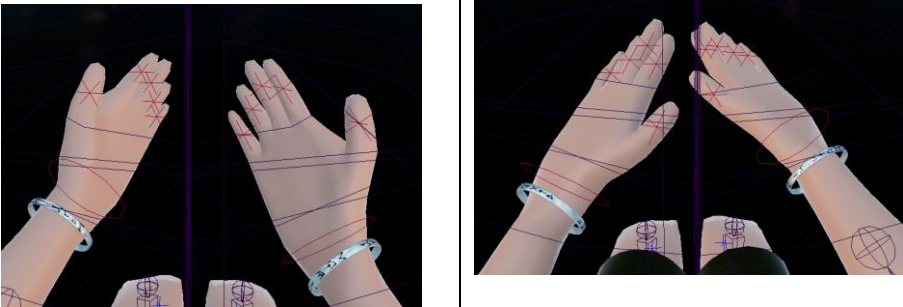
			continuous motion.
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9.2.2 Rotation

Bunk scene

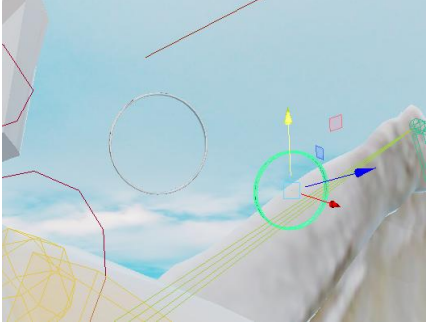
Fan rotation		What: Continuous rotation of the inner blade part of the fan. Why: To add realism and detail to the environment, especially in indoor scenes like the bunk, enhancing immersion. How: The fan blades were animated using keyframes or set-driven expression to rotate on the Y-axis in a looping motion, simulating a powered fan.
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Angel scene

Hand and ring rotation		What: Animation of the character's hand raising and rotating to view the bracelet. Why: To emphasise the transformation moment and draw attention to the magical bracelet, showing the character's reaction and curiosity. How: Animated using keyframes by rotating and lifting the arm object and wrist slightly. The ring itself also rotates slowly to add a magical effect while the character observes it.
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9.2.3 Scaling

Angel scene

Ring Transform		What: Transformation of the necklace into two bracelets through scaling and movement. Why: To visually represent the magical gift given by the angel, symbolising the main character's inner potential being unlocked. How: The necklace object was animated using scaling. It grows and moves onto each wrist using keyframes.
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9.2.4 Camera

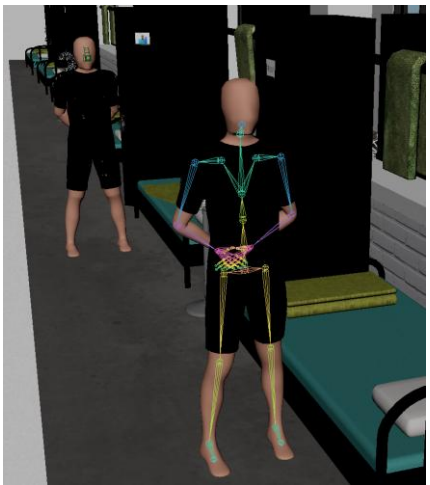
Bunk scene

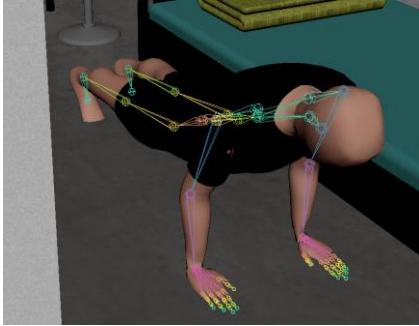
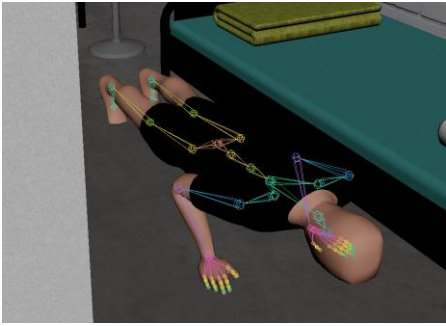
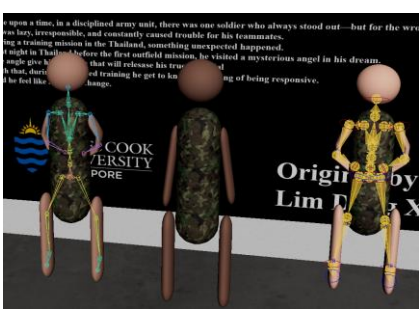
Part 1 camera		What: Camera movement starts with a wide shot of the bunk and slowly zooms in to focus on the character and the floating ring. Why: To establish the environment and build anticipation, guiding the viewer's attention from the setting to the magical event. How: Animated the camera with smooth zoom and movement keyframes.
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FPV camera movement			What: First-person perspective camera movement following the main character through most scenes. Why: To immerse the viewer directly into the character's experience, creating a personal and engaging viewpoint. How: The Camera is following the character's head and animated to mimic the eye of the character.
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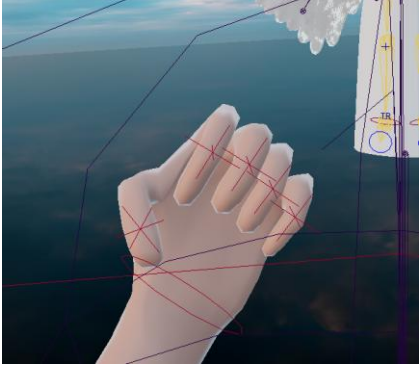
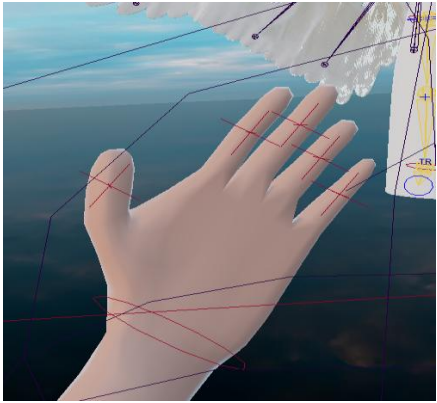
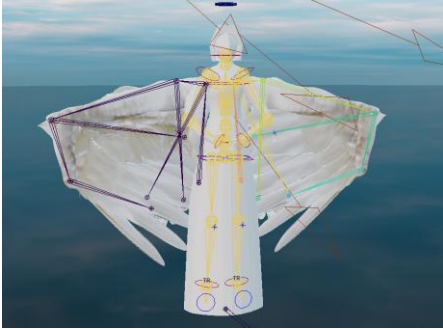
9.2.5 Bone

Bunk scene

Lim and Jackie's bone movement			What: Skeletal (bone) animation controlling the characters' body movements, including walking, gestures, and expressions. Why: To create realistic and natural character motions that bring Lim and Jackie to life and support storytelling. How: Used Maya's HumanIK rig system to animate bones with keyframes.
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Push up, punish bone movement.			What: Skeletal animation showing the main character performing push-ups. Why: Added multiple push-ups as punishment given by the commander. How: Animated the character's bones using keyframes in Maya's HumanIK rig, focusing on the realistic arm and body to simulate proper push-up form and effort.
Sergeant and commander bone movement			What: Skeletal animation of the Sergeant and Commander characters, including gestures, walking, and commanding poses. Why: To convey authority, emotion, and interaction with other characters, enhancing the storytelling. How: Animated using Maya's HumanIK rig system with keyframes, focusing on body language.

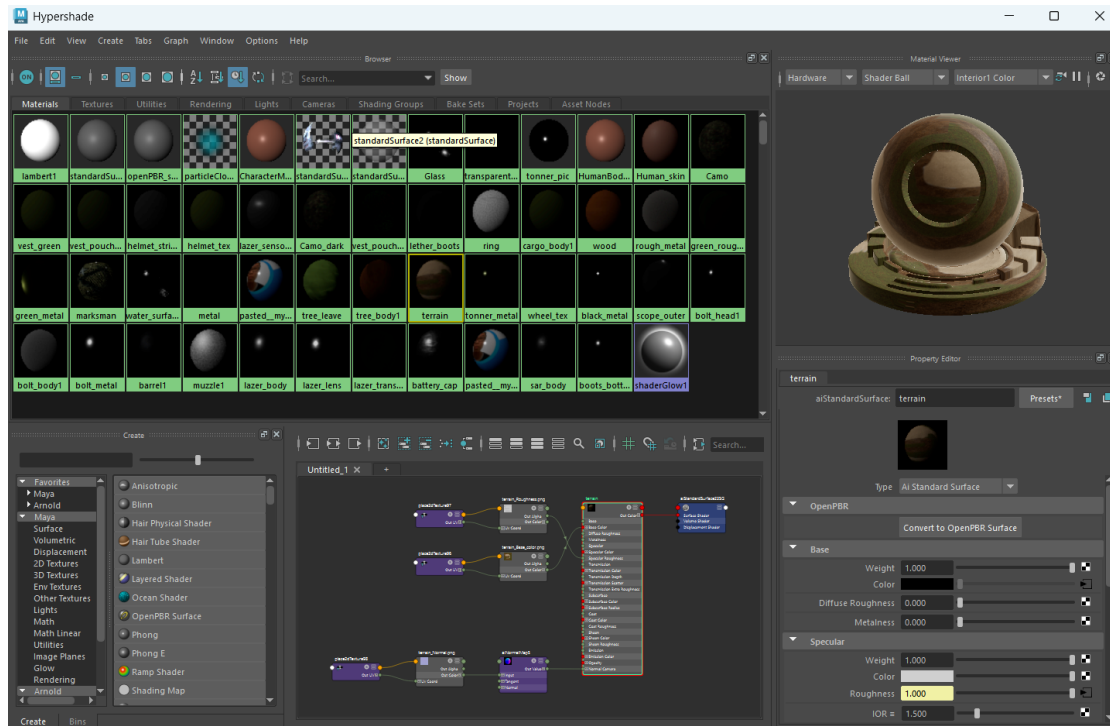
Angel scene

Character finger movement	 	What: Animation of the character's fingers opening and closing. Why: To add realism and expressiveness to hand movements. How: Keyframed the finger joints in Maya's rig, carefully adjusting rotations to create smooth and natural finger curling.
Angel and wing bone movement	 	What: Skeletal animation of the angel character's body and wings, including subtle wing flapping and body gestures. Why: To bring life to the angel, emphasising its divine and ethereal nature in the scene. How: Used Maya's HumanIK rig and custom bones for the wings, keyframing gentle flapping motions and fluid body movements to create a smooth animation.

9.3 Textures and Shaders

I created more texture for almost all the objects for the new object and scene using the same tools used in part 1.

I made all the textures using AI Standard Surface for Arnold to do rendering.



Texture Maps Used

- Base Colour: For the main surface colour.
- Normal Map: For surface details (scratches, cloth, skin bumps).
- Roughness: To control how shiny or matte the surface is.
- Metalness: For metallic parts like guns or wires.
- Emission Map: For glowing parts like the angel ring or ceiling light tube.

I followed a tutorial that uses the texture itself to create an animated water surface.



9.4 UV Map

What:

UV mapping is the process of unwrapping a 3D model's surface and projecting it into a 2D space so that textures can be accurately applied to the model.

Why:

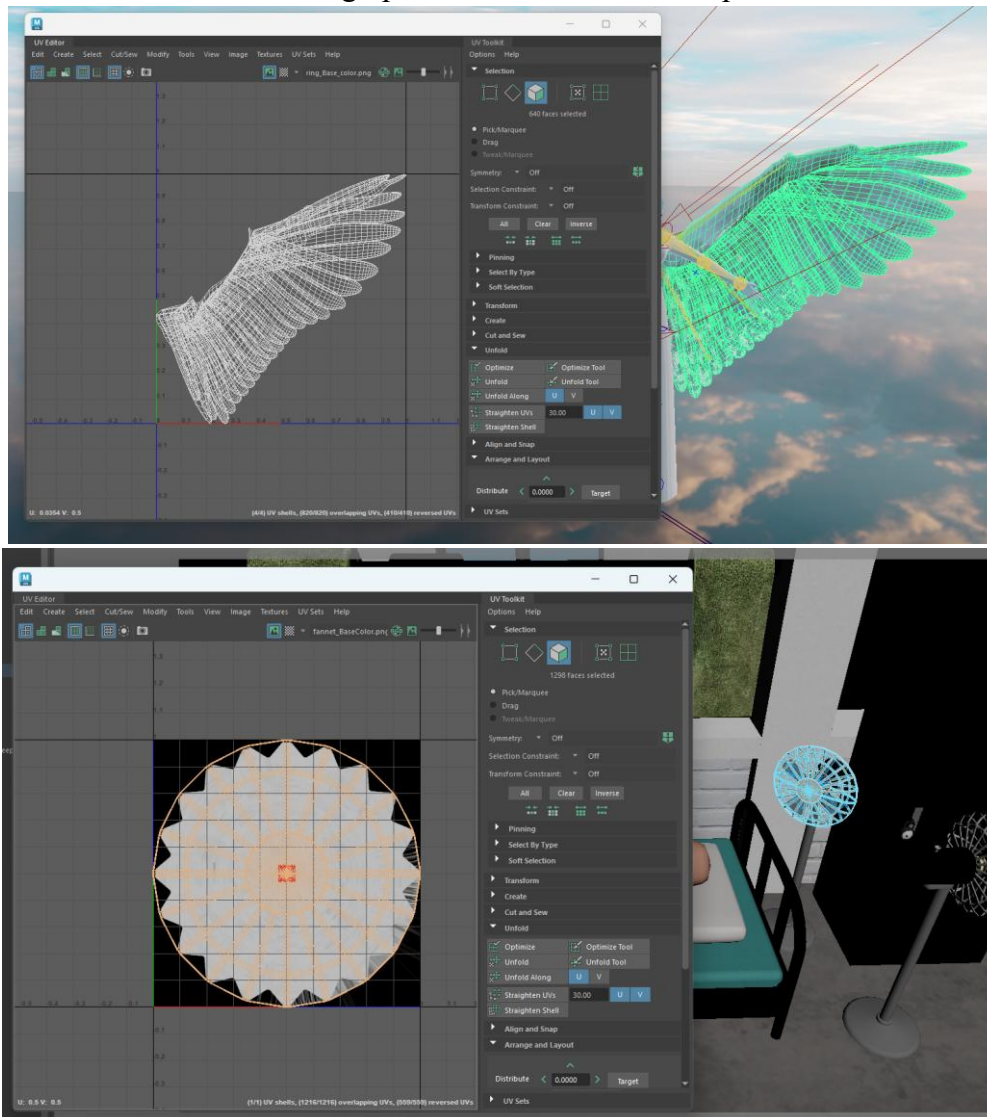
Without proper UV mapping, textures can appear stretched, misplaced, or distorted. UVs are necessary for applying image textures (like camouflage, wood, or skin detail) cleanly onto the surface of 3D models.

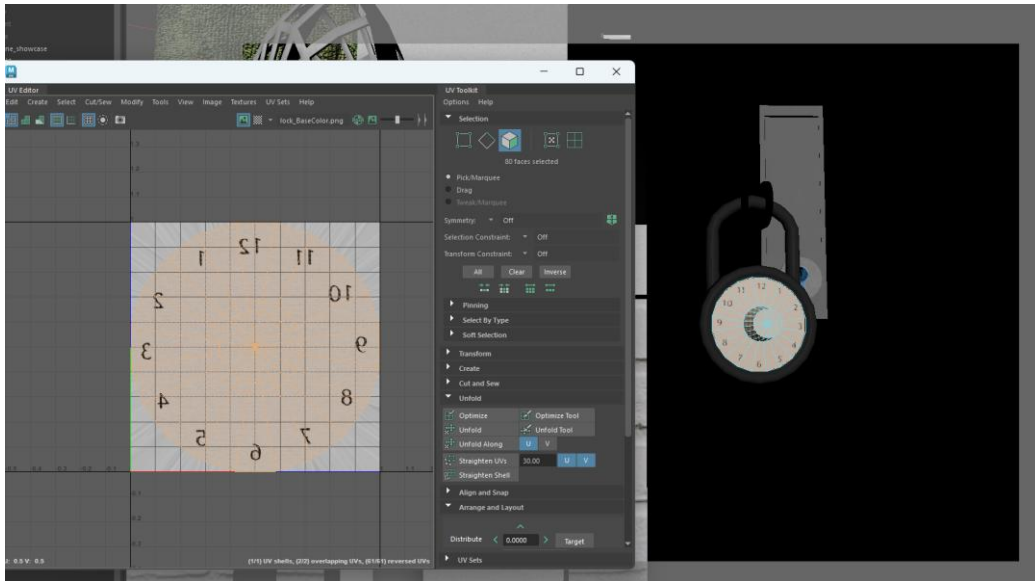
How:

I modified at least 70% UV maps for the texturing. Most of it uses the same step. First, I deleted the original UV map, as most steps will break the UV map. Then I created an orthographic camera-based UV map, some of the model. I stopped here as the camera-based UV map is enough. If not, I will start cutting the edge and unwarping it, then I will rotate it and straighten UVs, and finally lay it out and place it in the place I wanted it.

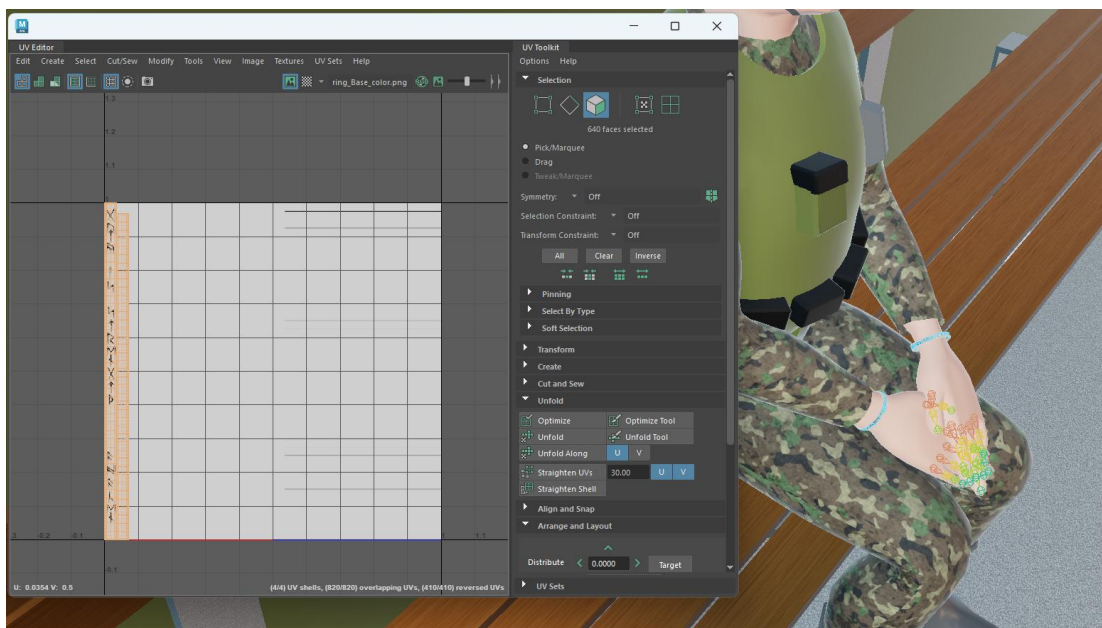
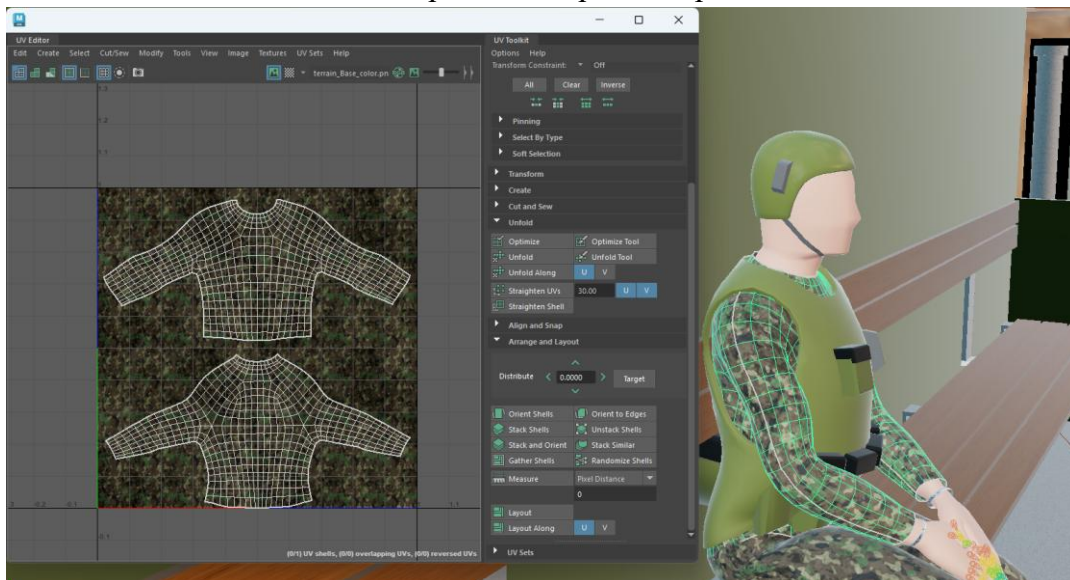
Some UV example:

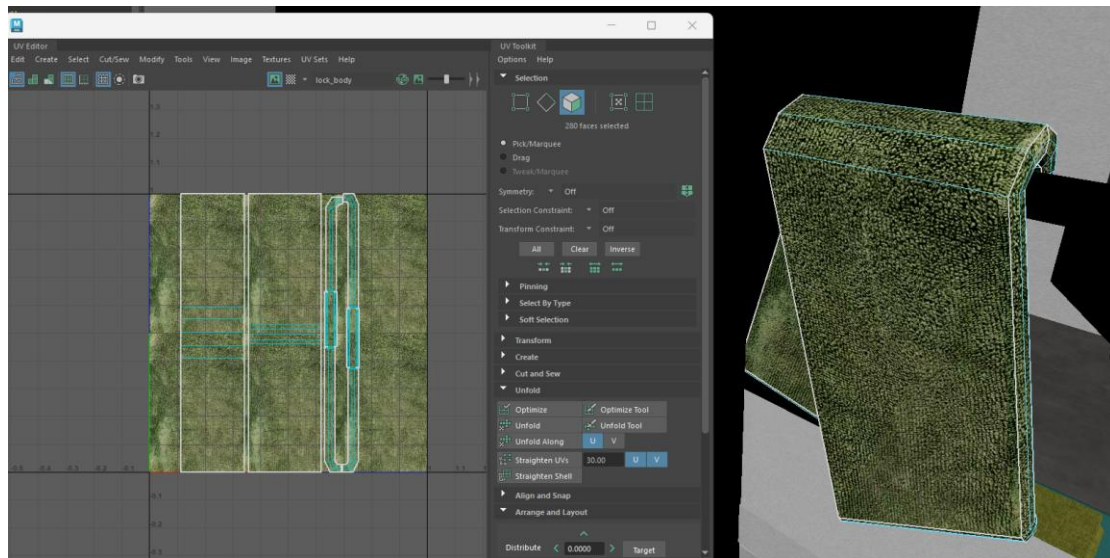
Orthographic camera-based UV map



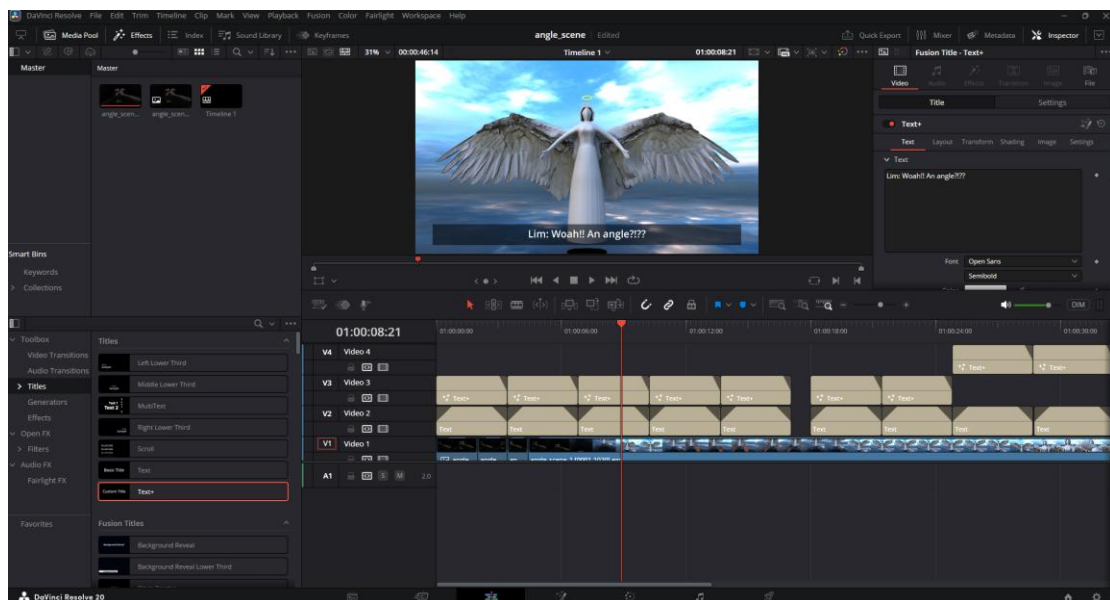


Complete unwarp UV map.





9.5 Added subtitle for the recording scene



9.6 Video:

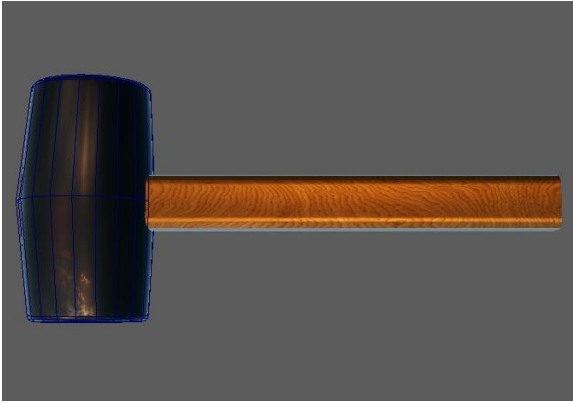
Part 2 required video (backup link): https://myjucedu-my.sharepoint.com/:v:/g/personal/dongxian_lim_my_jcu_edu_au/EfNqmKqVek1O1Otmh0x1fgBsHgwD_Wjf7MKNeBeNIAL6w?e=WB8ObQ

My finger rigging progress video link: https://myjucedu-my.sharepoint.com/:v:/g/personal/dongxian_lim_my_jcu_edu_au/Eb079fcVXWZKubDs6uVR1GoB_kNTPLUozqUiRBD3-gx9Q?e=HLgEOI whaw

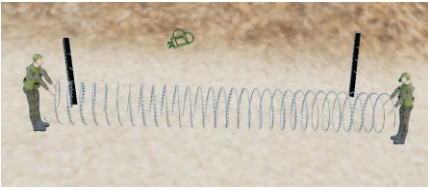
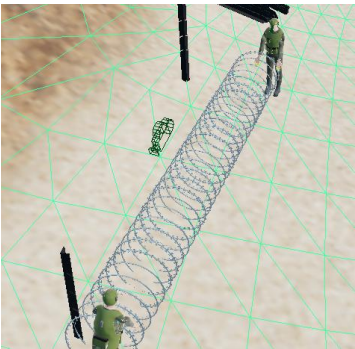
10. Part 3

10.1 Update content

10.1.1 Update Item

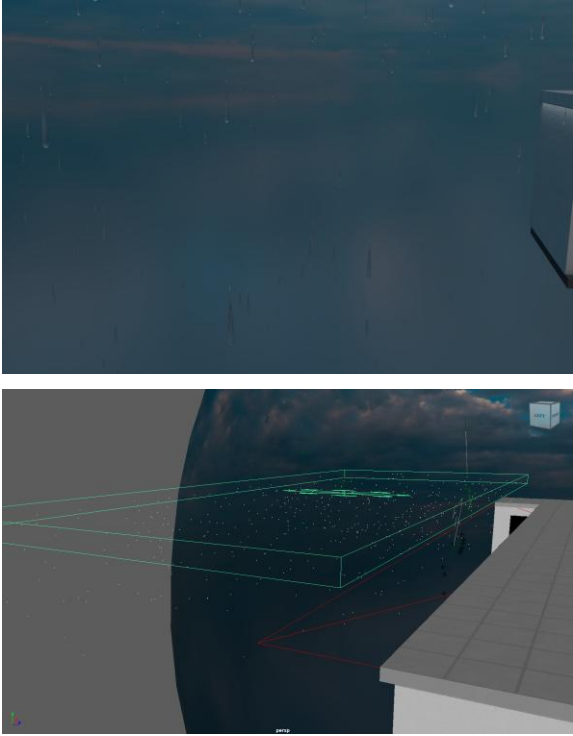
Hammer		What: A simple two-sided hammer used for securing equipment. Why: To hammer in the iron pole to stable the thorn wire. To show that an iron pole needs quite a power to be plugged into the ground. How: Handle created from an extended cube with bevelled edges to make it smooth. The head is made from a cylinder with an added edge loop in the middle and scaled to form a realistic but simple hammer head.
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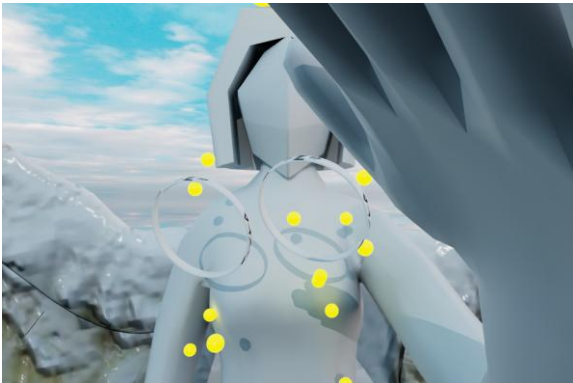
10.1.2 Update Animation

Thorn Wire extent (Scaling)		 	What: Increase the length of the thorn wire to bring it back to its original length. Why: To simulate the wire being pulled and extended during setup in the scene. How: Used scaling transformation on the thorn wire model over time to create the effect
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Iron Bar (Movement)			of it stretching. What: Animation of an iron bar being hammered into the ground. Why: To realistically show the process of securing the thorn wire in place. How: Keyframed vertical movement combined with timing to match the hammering action.
			What: Animation of a hammer swinging down to hit the iron bar. Why: To show the action of driving the iron bar into the ground for realism and scene continuity. How: Used keyframes to animate the hammer's rotation and position, syncing the motion with the iron bar's movement and impact timing.

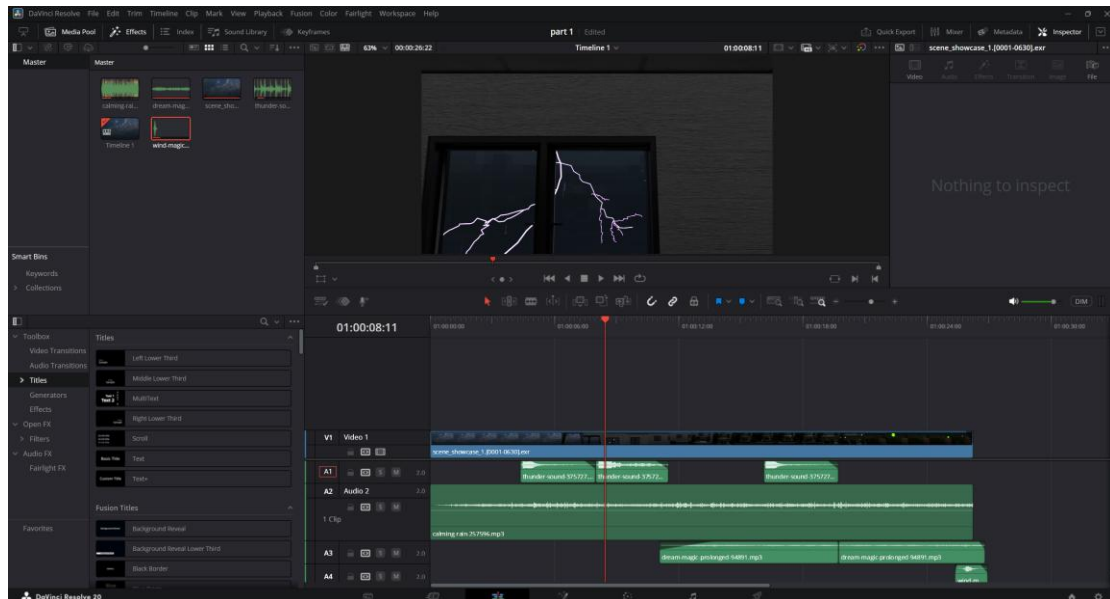
10.2 Particle Effects

Rain Particles		What: Rain particle effect using Maya's nParticles. Why: To simulate realistic rainfall for the indoor scene, enhancing the atmosphere. How: Created a volume nParticle emitter. Adjusting the variable. Since Maya's streak render type can't be rendered in Arnold, I created a raindrop model and instanced it to the particles. This way, each particle uses my custom raindrop model. A simple water-like texture was applied to make the rain look more realistic in Arnold render.
Thunder effects		What: Maya builds in effect with branching bolts and glow. Why: To simulate a thunderstorm environment alongside the rain particles, enhancing the atmosphere and realism of the rainy scene. How: I used Maya's built-in thunder effect between multiple locators to create branching lightning. I adjusted the

		thickness and added emission with a purple glow to simulate realistic thunder visuals.
Glowing sphere particles	 	What: Glowing floating sphere particles used in the bunk and angel scenes. Why: To create a magical or supernatural atmosphere in both scenes, enhancing the visual storytelling. How: Created using nParticles with spherical shape, upward movement, and glow effect applied through emission shader for a soft, magical appearance.
Glowing ring particles		What: Glowing ring particles used in the angel scene during spell-casting. Why: To visually represent magical energy being cast by the angel, enhancing the fantasy feel of the scene. How: Created using directional nParticles moving upward and instanced with a ring-shaped model. The yellow emission shader is applied to give a glowing, mystical effect.

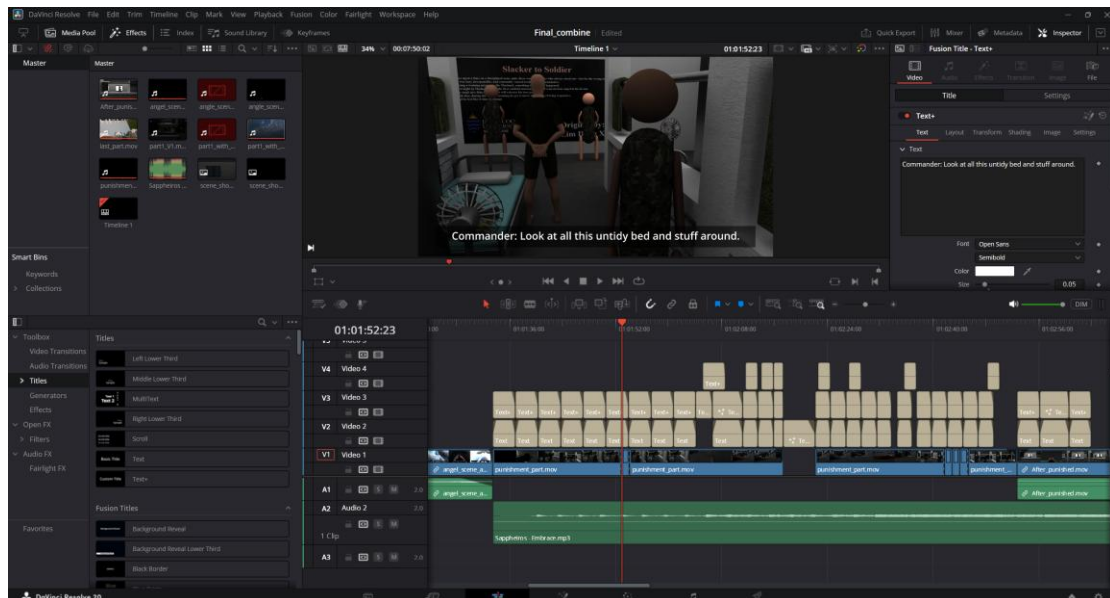
10.3 Audio Effects

Audio and Sound Effect:



To enhance the final animation, I added background sounds and effects such as thunder, rain, and magic using DaVinci Resolve. The sound effects were downloaded for free from Pixabay. These audio elements help create a more immersive atmosphere and match the visual effects in the animation.

10.4 Subtitles



Subtitles are added to show which character is speaking, helping the audience better follow the story. This improves clarity, especially in scenes with multiple characters. It also enhances the overall viewing experience by making the animation more immersive and easier to understand.

10.6 Video

Final video backup link:

- YouTube link: <https://youtu.be/XD7OLTQihcM>
- My OneDrive: [Final_combination.mp4](#)

Progress Video:

- Thunder and Rain particles Creation (backup OneDrive link):
[rain_thunder.mp4](#)

Reference

Software/Tools:

3D Substance Painter: <https://www.adobe.com/sg/products/substance3d.html>
Actor Core: <https://actorcore.reallusion.com/auto-rig/accurig>
Davinci Resolve: <https://www.blackmagicdesign.com/products/davinciresolve>
MAYA download: <https://www.autodesk.com/sg/products/maya/overview>
Normal Mapping Online: <https://cpetry.github.io/NormalMap-Online/>
Photoshop: <https://www.adobe.com/sg/products/photoshop.html>

Tutorial video:

Academic Phoenix Plus. (2023, May 3). Quickly rig a character in AccuRig and Maya 2023 HumanIK [Video]. YouTube.
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Alex Cheparev. (2022, December 8). Easy 3D character modeling in Maya - Part 2 - Hands & feet [Video]. YouTube.
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Texture/ Image texture:

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- SoundReality. (n.d.). Thunder Sound. Pixabay. <https://pixabay.com/sound-effects/thunder-sound-375727/>
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